

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410925
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Wesley; Mitchell J. et al.

Air purifier and air purifier accessory

Abstract

An air purifier and air purifier accessory for inside air. The air purifier includes a housing and an air outlet. The air purifier accessory is disposed on the air outlet. The air purifier accessory includes a frame containing a filter media, the frame adapted to be mounted at least partially over the air outlet of the air purifier to filter air discharged from the air purifier.

Inventors: Wesley; Mitchell J. (Grand Rapids, MI), Smith; Derek (Greenville, MI), Baydoun; Habib (Dearborn Heights, MI)

Applicant: BISSELL Inc. (Grand Rapids, MI)

Family ID: 87070338

Assignee: BISSELL Inc. (Grand Rapids, MI)

Appl. No.: 18/095548

Filed: January 11, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230221019 A1	Jul. 13, 2023

Related U.S. Application Data

us-provisional-application US 63308640 20220210
us-provisional-application US 63298312 20220111

Publication Classification

Int. Cl.: F24F8/10 (20210101); F24F8/108 (20210101); F24F13/06 (20060101); F24F13/20 (20060101); B01D46/00 (20220101); B01D46/24 (20060101)

U.S. Cl.:

CPC **F24F8/108** (20210101); **F24F13/06** (20130101); **F24F13/20** (20130101); B01D46/0005 (20130101); B01D46/0047 (20130101); B01D46/2411 (20130101); F24F2013/0616 (20130101)

Field of Classification Search

CPC: F24F (8/108); F24F (13/06); F24F (13/20); F24F (2013/0616); F24F (8/10); F24F (8/80); F24F (13/28); B01D (46/0005); B01D (46/0047); B01D (46/2411)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5753000	12/1997	Chiu	55/357	F24F 8/108
6053968	12/1999	Miller	96/16	F24F 8/22
9579597	12/2016	Gruenbacher	N/A	B01D 46/02
10926210	12/2020	Aries	N/A	F24F 13/06
11717782	12/2022	Oh	55/467	F04D 29/626
12134072	12/2023	Hoyda	N/A	B01D 53/346
2014/0150658	12/2013	Prax	96/74	B01D 46/0002
2014/0331859	12/2013	Gruenbacher	95/26	F24F 13/28
2015/0114233	12/2014	Lee	261/114.5	F24F 6/06
2017/0122596	12/2016	Lee	N/A	F24F 11/30
2017/0122602	12/2016	Son	N/A	F24F 6/04
2017/0138616	12/2016	Wang	N/A	F24F 1/029
2017/0153040	12/2016	Wang	N/A	F24F 13/28
2019/0107302	12/2018	Liu	N/A	F24F 8/10
2019/0137124	12/2018	Mcgarva	N/A	F24F 13/0272
2019/0301756	12/2018	Luo	N/A	F24F 8/30
2019/0308122	12/2018	Aries	N/A	F24F 13/06
2020/0158357	12/2019	Son	N/A	F04D 25/166
2020/0224892	12/2019	Yang	N/A	F24F 1/029
2020/0289968	12/2019	Scholten	N/A	B01D 46/0009
2020/0298157	12/2019	Kang	N/A	B01D 46/4227
2020/0298160	12/2019	Jeon	N/A	B01D 46/58
2020/0298161	12/2019	Jeon	N/A	B01D 46/0008
2020/0298167	12/2019	Jeon	N/A	B01D 46/0013
2020/0298168	12/2019	Lee	N/A	B01D 46/46
2020/0300485	12/2019	Kang	N/A	B01D 46/4245
2020/0300502	12/2019	Kang	N/A	F24F 13/20
2020/0355191	12/2019	Park	N/A	F04D 17/16
2021/0086219	12/2020	Hsiao	N/A	B08B 3/08
2021/0172626	12/2020	Jung	N/A	F04D 25/08
2021/0278097	12/2020	Min	N/A	F24F 13/20
2021/0325057	12/2020	Gong	N/A	F24F 3/16
2021/0372640	12/2020	Yang	N/A	F24F 6/04
2021/0381518	12/2020	Horton	N/A	F04D 25/10

2022/0011009	12/2021	Lee	N/A	F24F 13/20
2022/0011011	12/2021	Yang	N/A	F24F 13/082
2022/0023789	12/2021	Oh	N/A	B01D 46/0002
2022/0032222	12/2021	Park	N/A	F24F 13/10
2022/0032224	12/2021	An	N/A	B01D 46/0005
2022/0184542	12/2021	Kim	N/A	F24F 8/108
2022/0184543	12/2021	Choi	N/A	B01D 46/0049
2022/0235956	12/2021	Huang	N/A	F24F 13/20
2022/0275955	12/2021	Dunberger	N/A	F24F 13/20
2023/0071996	12/2022	Yang	N/A	F24F 8/108
2023/0073213	12/2022	Yang	N/A	A61L 9/16
2023/0152000	12/2022	Ha	95/273	F24F 13/28
2023/0158435	12/2022	Ha	55/411	F24F 8/108
2023/0366571	12/2022	Smith	N/A	B01D 46/103
2023/0381704	12/2022	Smith	N/A	B01D 46/0047
2024/0017195	12/2023	Mani	N/A	F24F 8/80
2024/0017199	12/2023	Lee	N/A	F24F 8/10
2024/0019139	12/2023	Wesley	N/A	B01D 46/0027
2024/0108773	12/2023	Kim	N/A	G05D 1/246
2024/0216863	12/2023	Liu	N/A	B01D 53/869
2024/0426489	12/2023	Chen	N/A	F24F 13/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
205570092	12/2015	CN	N/A
107388405	12/2016	CN	F24F 8/10
109140620	12/2018	CN	F24F 8/10
109173600	12/2018	CN	B01D 50/60
109595712	12/2018	CN	F24F 11/32
112325438	12/2020	CN	F24F 13/28
116045434	12/2022	CN	F24F 8/80

Primary Examiner: Hobson; Stephen

Attorney, Agent or Firm: Warner Norcross + Judd LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of U.S. Provisional Patent Application No. 63/298,312, filed Jan. 11, 2022, and U.S. Provisional Patent Application No. 63/308,640, filed Feb. 10, 2022, both of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Air purifying devices filter particles from inside air and/or kill airborne pathogens. Such devices typically use filters or ion injectors to purify the air surrounding the device. Yet there are continual issues in operation, function, and ornamental appearance that remain unresolved.

(2) Current air purifiers often lack an amount of filter media sufficient to remove odors from the air. This is especially true for smaller air purifying units that have minimal space in the interior of the housing. These units often have combined media types stored in the interior of the unit.

Typically, the filter media is on the inlet side of the air flowing through the interior of the air purifier housing. Because of the limited space, a limited amount of odor absorbing and/or eliminating filter media can be used in conjunction with HEPA filters. For example, often, odor removing media is in a mesh form or it is woven into the HEPA media in trace amounts, neither of which is significant enough to sufficiently capture, absorb, or remove odor causing volatile organic compounds (VOCs), especially in a concentrated odor environment, such as in pet areas or near diaper pails. Typically, granules or pellets of odor removing media function optimally in this environment, however, using granules or pellets on the interior of the housing significantly reduces the airflow through the air purifier and thus, not enough air is being cleaned and/or purified through the unit to remove the odors.

BRIEF DESCRIPTION

- (3) An air purifier and air purifier accessory with improved ability to capture, absorb, and/or remove odor causing volatile organic compounds (VOCs) from inside air is provided herein. Aspects of the disclosure relate to an improved outlet or exhaust accessory that is adapted to be mounted on the air outlet of an air purifier to remove odors from the air.
- (4) An air purifier accessory includes frame containing a filter media, the frame adapted to be mounted at least partially over the air outlet of the air purifier.
- (5) In one aspect of the disclosure, an air purifier assembly includes an air purifier and an air purifier accessory. The air purifier includes a housing having a longitudinal axis, an air flow path through the housing including an air inlet and an air outlet, a blower disposed within the housing, the blower defining a portion of the air flow path and adapted to pull air in through the air inlet and to push air outlet out of the air outlet, and an air treatment device disposed within the housing to treat air flowing through the air flow path. The air purifier accessory, which can be disposed on the air outlet, outside of the housing, to treat air flowing out of the air outlet, includes a frame, a filter media within the frame for filtering air discharged from the air purifier, at least one air inlet on an inflow side of the frame, a plurality of air outlets on an outflow side of the frame, wherein the outflow side of the frame is perpendicular to the inflow side of the frame, and an air deflector configured to deflect the air to be filtered as it flows through the frame, the air deflector defining a closed area within the frame in which air does not flow and an open area within the frame in which air flows, wherein air exiting the air purifier flows axially into the open area through the inflow side of the frame and flows radially out of the open area through the outflow side of the frame.
- (6) In another aspect of the disclosure, an air purifier accessory includes a frame adapted to be coupled to an exterior of an air purifier and defining an longitudinal axis, a filter media within the frame for filtering air discharged from the air purifier, at least one air inlet on an inflow side of the frame, a plurality of air outlets on an outflow side of the frame, wherein the outflow side of the frame is perpendicular to the inflow side of the frame, and an air deflector configured to deflect the air to be filtered as it flows through the frame, the air deflector defining an open area within the frame in which air flows, wherein air exiting the air purifier flows axially into the open area through the inflow side of the frame and flows radially out of the open area through the outflow side of the frame.
- (7) These and other features and advantages of the present disclosure will become apparent from the following description of particular embodiments, when viewed in accordance with the accompanying drawings and appended claims.
- (8) Before the embodiments of the invention are explained in detail, it is to be understood that the invention is not limited to the details of operation or to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention may be implemented in various other embodiments and of being practiced or being carried out in alternative ways not expressly disclosed herein. In addition, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including” and “comprising” and variations thereof is meant to

encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof. Further, enumeration may be used in the description of various embodiments. Unless otherwise expressly stated, the use of enumeration should not be construed as limiting the invention to any specific order or number of components. Nor should the use of enumeration be construed as excluding from the scope of the invention any additional steps or components that might be combined with or into the enumerated steps or components. Any reference to claim elements as “at least one of X, Y and Z” is meant to include any one of X, Y or Z individually, and any combination of X, Y and Z, for example, X, Y, Z; X, Y; X, Z; and Y, Z.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the drawings:
- (2) FIG. 1 is a perspective view of an air purifier with an air purifier accessory according to various aspects described herein;
- (3) FIG. 2 is a perspective view of the air purifier of FIG. 1, with the air purifier accessory exploded from a housing of the air purifier;
- (4) FIG. 3 is an exploded view of the air purifier accessory of FIG. 1;
- (5) FIG. 4 is a top view of the air purifier accessory of FIG. 1;
- (6) FIG. 5 is a bottom view of the air purifier accessory of FIG. 1;
- (7) FIG. 6 is a cross-sectional view through the air purifier accessory of FIG. 1;
- (8) FIG. 7 is a cross-sectional view through an upper portion of an air purifier an air purifier accessory according to another aspect of the disclosure;
- (9) FIG. 8 is an exploded view of an air purifier accessory according to yet another aspect of the disclosure;
- (10) FIG. 8A is a cross-sectional view through an upper portion of the air purifier of FIG. 1 with the air purifier accessory of FIG. 8;
- (11) FIG. 9 is a top perspective view of the air purifier accessory of FIG. 8;
- (12) FIG. 10 is a bottom perspective view of the air purifier accessory of FIG. 8;
- (13) FIG. 11 is an exploded view of an air purifier accessory according to still another aspect of the disclosure;
- (14) FIG. 12 is a top perspective view of the air purifier accessory of FIG. 11;
- (15) FIG. 13 is a bottom perspective view of the air purifier accessory of FIG. 11;
- (16) FIG. 14 is an exploded view of an air purifier accessory according to a further aspect of the disclosure;
- (17) FIG. 15 is a top perspective view of the air purifier accessory of FIG. 14;
- (18) FIG. 16 is a bottom perspective view of the air purifier accessory of FIG. 14;
- (19) FIG. 17 is a top perspective view of an air purifier accessory according to yet a further aspect of the disclosure;
- (20) FIG. 18 is a bottom perspective view of the air purifier accessory of FIG. 17;
- (21) FIG. 19 is a perspective view of the air purifier from FIG. 1-2, with an air purifier accessory according to still a further aspect of the disclosure, and with the air purifier accessory exploded from the housing of the air purifier;
- (22) FIG. 20 is a cross-sectional view through the air purifier and accessory of FIG. 19, with the accessory installed on the air purifier;
- (23) FIG. 21 is an exploded view of the air purifier accessory of FIG. 19;
- (24) FIG. 22 is a bottom perspective view of the air purifier accessory of FIG. 19; and
- (25) FIG. 23 is a cross-sectional view through an upper portion of the air purifier from FIG. 1-2, with an air purifier accessory according to a further aspect of the disclosure.

DETAILED DESCRIPTION

(26) Aspects of the disclosure relate to an air purifier and/or an air purifier accessory. While primarily discussed herein in terms of an air purifier for indoor air, aspects of the filter accessory and embodiments thereof disclosed herein are applicable to other types of air filtration apparatus.

(27) The term “clean” as used herein is to describe the removal of pollutants from air as compared to the ambient air. Pollutants can include dirt, dust, volatile organic compounds (VOCs), biological contaminants (e.g., bacteria, viruses, mold spores, waste products, etc.), soot particles, and any other pollutants that can be found in indoor and/or outdoor flows.

(28) The term “purify” is used herein to describe killing or inactivating biological contaminants, pathogens, or microorganisms in the ambient air. It is noted that the air purifier apparatus can have a variety of applications including commercial or home based application.

(29) As used herein, the term “upstream” refers to a direction that is opposite the air flow direction, and the term “downstream” refers to a direction that is in the same direction as the air flow.

Additionally, as used herein, the terms “radial” or “radially” refer to a direction away from a common center. Furthermore, as used herein, the term “set” or a “set” of elements can be any number of elements, including only one.

(30) All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, secured, fastened, connected, and joined) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order, and relative sizes reflected in the drawings attached hereto can vary.

(31) FIGS. 1-2 illustrate an air purifier **10** including a housing **12** having a longitudinal axis X. An air flow path through the housing **12** includes an air inlet **27** and an air outlet **30**. An air treatment device **38** is disposed within the housing **12** to filter, clean, and/or purify air flowing through the air flow path. A blower **31** can define a portion of the air flow path and is disposed within the housing **12**. An air purifier accessory **40** is disposed on the air outlet **30**, outside of the housing **12**, to filter, clean, and/or purify air flowing out of the air outlet **30**. The air purifier accessory **40** is therefore disposed downstream of the air treatment device **38**, blower **31**, and air outlet **30**. The air purifier **10** and accessory **40** can form an assembly.

(32) An interior **14** of the air purifier **10** can at least partially be defined by the housing **12**. The housing **12** defines an exterior periphery **16** for the air purifier **10**. The housing **12** can further define an upper portion **18** and a lower portion **20**. The upper portion **18** and the lower portion **20** can divide the housing **12** in half vertically. However, it is contemplated that the upper portion **18** can be greater than or less than half of the housing **12**, with the lower portion **20** defining the remaining portion of the housing **12**. The lower portion **20** can be adapted to rest on a surface, with the upper portion **18** spaced from the surface on which the lower portion **20** rests. The lower portion **20** can include a base **22**. Optionally, the base **22** can include legs **24** or other supporting structures.

(33) The air treatment device **38** and blower **31** are disposed within the interior **14** of the air purifier **14**. By way of non-limiting example, the air treatment device **38** can include a filter, an ion injector, or any combination thereof, and including multiples thereof. By way of non-limiting example, the blower **31** can include a fan and a motor to drive the fan. While the motor is energized, air is pulled in through the air inlet **27** and pushed out of the air outlet **30** by the fan.

(34) In the embodiment shown, the blower **31** is disposed in the upper portion **18** of the housing

and the air treatment device **38** is disposed in the lower portion **20** of the housing **12**, generally below the blower **31**. Other spatial arrangements are possible. The blower **31** is preferably disposed downstream of the air treatment device **38** within the air flow path, regardless of the spatial arrangement of the components.

(35) The air inlet **27** can be provided in various locations and configurations on the housing **12**. By way of non-limiting example, the air inlet **27** can include a plurality of perforations **29** in the lower portion **20** of the housing **12**. The plurality of perforations **29** can be included on one or any number of sides of the housing **12**. The plurality of perforations **29** can be generally uniform in size and shape, or may have a variety of sizes and shapes.

(36) The housing **12** can be a monolithic body, e.g., a single body that is a single, non-separable piece, or formed as a single unitary piece at manufacture, as opposed to being formed by combining separate elements into one during manufacture. Alternatively, the housing **12** can be defined by more than one component coupled together. The shape of the housing **12**, as illustrated by way of example is a cylinder having a base or cross-sectional shape that is a circle, however it is contemplated that the housing **12** can be any shape, such as, but not limited to, a prism, having a base or cross-sectional shape that is a squircle (e.g., a shape intermediate between a square and a circle), a round rectangular prism having a base or cross-sectional shape that is a rounded rectangle or a square, or an elliptic cylinder having a cross-sectional shape that is an ellipse or oval.

(37) A top portion **26** of the housing **12** is illustrated as being provided adjacent the upper portion **18**. A lip or ledge **28** can define a perimeter of the top portion **26**. The ledge **28** can be uniformly formed with the housing **12** or be a separate component of the housing **12**. In the illustrated example, and by way of non-limiting example the top portion **26** has a similar shape to a cross-section of the housing **12**. That is, the top portion **26** is generally shaped as a circle, although this need not be the case. That is, the housing **12** can have a cross-section similar in shape to the top portion **26**; alternatively, the top portion **26** can have a different shape or be a different size than a cross-section of the housing **12**.

(38) The ledge **28** can generally shaped as a circle, or can otherwise have a cross-section similar in shape to a cross-section of the housing **12**. Alternatively, the ledge **28** can have a different shape or be a different size than a cross-section of the housing **12**.

(39) The air outlet **30** can be provided in various locations and configurations on the housing **12**. By way of non-limiting example, the air outlet **30** can be provided in within the top portion **26** of the housing **12**, and is shown herein as a vent. The vent can include a plurality of flow diverters **32**, which define a plurality of openings **34** therebetween. While illustrated generally as radially extending vanes, the flow diverters **32** can be any shape, size, profile, spacing, or orientation. Further, the flow diverters **32** can be defined by two or more pieces or components. Other configurations for the air outlet **30** are possible.

(40) A user interface **36** can be provided with the top portion **26** of the housing **12**. More specifically, the user interface **36** can be located at an upper surface of the air purifier **10**, although this need not be the case and the user interface **36** can be provided at any suitable location. It is contemplated, however, that the user interface **36** can be located on one or more sides of the housing **12**. In the non-limiting illustrated example, the user interface **36** can be operably coupled to the housing **12**. This can include, but is not limited to, being mounted to or formed with the housing **12**. Further still, while a single user interface **36** is shown, it is further contemplated that the air purifier **10** can include more than one user interface. Optionally, the air purifier **10** can be remotely controlled by one or more electronic devices such as, but not limited to, a handheld remote or a cellular phone. The user interface **36** can include or be couple to a controller (not shown). The user interface **36** can include various features including, but not limited to, indicator lights providing information about the air purifier **10** to the user and/or buttons that adjust one or more settings or monitor the status of one or more components of the air purifier **10**.

(41) The air purifier **10** can be powered by an external power supply, such as a household power

supply, or by an internal power supply, such as a battery pack.

(42) The air purifier **10** includes an air purifier accessory **40** on the air outlet **30**, outside of the air purifier housing **12**. The air purifier accessory **40** avoids the space and airflow constraints of prior configurations. The accessory **40** is positioned directly on the air outlet **30** of the purifier **10**. Once the filter media is fully saturated, the accessory **40** can be discarded and replaced. In one aspect, the accessory **40** is a disposable consumable article.

(43) The air purifier accessory **40** can be configured to contain any amount and/or type of filter media required for a specific odor rich area. For example, the air purifier accessory **40** can contain activated carbon pellets used to adsorb VOCs and mixed with Zeolites specific to ammonia removal to more effectively remove cat litter odors from a litter box area.

(44) In some embodiments, the air purifier accessory **40** is removable from the air purifier **10**. In being removable, i.e. not an integrated part of the purifier **10**, the accessory **40** can be removed and replaced as needed. For example, once the filter media is fully saturated, the accessory **40** can be removed for replacement of the saturated filter media with new filter media. Also, a user can customize the filter media inside the accessory **40** depending on the specific odor to address. Users benefit from both the effective odor removal and the customizability of the filter media to meet their odor removal requirements.

(45) As noted above, in some embodiments, the accessory **40** can be replenished with a new filter media. Alternatively, the accessory **40** can be configured as a disposable component, with the accessory **40** being disposed of when saturated and replaced with a new accessory. Where the accessory **40** or a portion thereof is configured for disposal after use, recyclable, biodegradable and/or compostable material, or combinations thereof, may be used to manufacture the accessory **40**.

(46) The air purifier accessory **40** is provided on the top portion **26** of the air purifier **10** and covers the air outlet **30** such that all air flowing out of the housing **12** passes through the air purifier accessory **40**. The air purifier accessory **40** can be removable, press fit, fastened, snap fit, or permanently affixed to the air purifier **10**.

(47) FIG. 3 is an exploded view of the air purifier accessory **40**. The air purifier accessory **40** comprises a frame including an upper housing **42** and a lower housing **44**. An upper mesh **46** is disposed between the upper housing **42** and the lower housing **44** and a lower mesh **48** is disposed beneath the lower housing **44**. The housings **42**, **44** may be plastic components manufactured via molding or other suitable method, and may be snapped, screwed, or otherwise secured together. The mesh **46**, **48** can be formed from a fine metal mesh, or from other suitable mesh.

(48) A filter media **50** is contained by the frame. The filter media **50** is configured to remove odors from the air being exhausted from the air purifier **10** via the outlet **30**, and may capture, absorb, and/or remove odor-causing VOCs. In certain embodiments, the filter media **50** is chosen from carbon, zeolite (understood to include at least zeolite X, zeolite Y, ultrastable zeolite Y, ZSM zeolite, offretite, and beta zeolite, among others), silica gel, faujasite, chabazite, clinoptilolite, mordenite, silicalite, metal organic frameworks, metal oxide, polymers, resins, and combinations thereof. In one embodiment, the filter media **50** is activated carbon pellets mixed with zeolites specific to ammonia removal.

(49) It is noted that in FIG. 3, only a portion of filter media **50** is shown for illustration purposes. The filter media **50** can substantially fill the open space in the lower housing **44** between the upper and lower mesh **46**, **48**. Alternatively, the filter media **50** can fill less than all of the open space, for example by being disposed on a radially interior portion or a radially exterior portion, in order to leave some open space unfilled.

(50) The lower housing **44** and/or lower mesh **48** can form an air inlet side or inflow side of the accessory **40** and the upper housing **42** and/or upper mesh **46** can form an air outlet side or outflow side of the accessory **40**.

(51) Each housing **42**, **44** and mesh **46**, **48** can have openings for air to pass with the housings **42**,

44 having larger openings and the mesh **46, 48** having finer openings. The mesh **46, 48** can prevent the odor media **50** from falling through the large openings of the housings **42, 44**.

(52) The housings **42, 44** can include a plurality of flow diverters **43A, 45A**, which define their respective openings **43B, 45B** therebetween. While illustrated generally as radially extending vanes, the flow diverters **43A, 45A** can be any shape, size, profile, spacing, or orientation. Further, the flow diverters **43A, 45A** can be defined by two or more pieces or components. Other configurations for the housings **42, 44** are possible.

(53) The accessory **40** can optionally include an access opening **52** in the center to allow access to the user interface **36** (FIG. 2). The access opening **52** can be formed by concentric openings in each of the housings **42, 44** and mesh **46, 48**, such that a center area of the accessory **40** is clear and the user interface **36** is accessible when the accessory **40** is placed on the air outlet **30** of the purifier **10**.

(54) The overall shape of accessory **40**, as illustrated by way of example is a cylinder having a base or cross-sectional shape that substantially matches that of the housing **12**, so that the accessory **40** is harmonious with the exterior periphery **16** of the purifier **10**. However, it is contemplated that the accessory **40** can be any shape, including shapes that differ from the shape of the housing **12**.

(55) FIG. 6 is a cross-sectional view through the air purifier accessory **40**, where the mesh and **46, 48**, and media **50** are not shown for clarity. The accessory **40** can have an extending portion **56** that is at least partially received in the top portion **26** of the air purifier **10** (FIG. 1), for example such that the extending portion **56** is disposed inwardly of the lip or ledge **28**. Yet other physical features for aligning the accessory **40** on the air purifier **10** are possible. In one non-limiting example, the alignment features can be configured to align the flow diverters **45A** on the outlet side of the accessory **40** with the flow diverters **32** of the air purifier outlet **30** to limit air flow restriction.

(56) FIG. 7 is a cross-sectional view through an air purifier **10A** and an air purifier accessory **40A** according to another aspect of the disclosure. The air purifier **10A** and air purifier accessory **40A** are substantially similar to those of FIGS. 1-7, and like elements are referred to with the same reference numeral bearing a letter "A." The air purifier **10A** differs by lacking a user interface located at an upper surface of the air purifier **10A**. Accordingly, the accessory **40A** does not need an access opening in the center, and can instead have a uniform surface **58** forming the upward facing surface of the assembled air purifier **10A** and accessory **40A**.

(57) To mount the accessory **40A** on the air purifier **10A**, the accessory **40A** can have a mounting flange **60** that extends downwardly from the lower housing **44A** that overlaps the top portion **26A** of the air purifier **10A**. As such, the accessory **40A** can be mounted without requiring any special structure on the air purifier **10A** and the accessory **40A** can be seated onto any complementary-shaped air purifier device. Alternatively, the accessory **40A** can have the same mounting structure (e.g., groove and extending portion) as shown in FIG. 6.

(58) FIGS. 8-10 show an air purifier accessory **40B** according to another aspect of the disclosure. The air purifier accessory **40B** is substantially similar to that of FIGS. 1-7, and like elements are referred to with the same reference numeral bearing a letter "B." The air purifier accessory **40B** includes a plurality of dividers **62** that create chambers **64** for the filter media **50B**. These chambers **64** serve to prevent damage to the filter media **50B** by minimizing shifting of the media **50B**. In one embodiment, the lower mesh **48B** can include the diverters **62**. For example, the dividers **62** can be formed with or otherwise attached to the lower mesh **48B**. The chambers **64** can further be defined by the upper mesh **46B** and the lower housing **44B**, with filter media **50B** substantially filling each of the chambers **64**.

(59) The accessory **40B** includes at least one air channel that allows air exiting the air purifier **10** (FIG. 1) to bypass the filter media **50B** so as to not suffocate the air purifier **10**. In the embodiment shown, an inner channel **66** and an outer channel **68** are provided and are disposed radially inwardly and radially outwardly, respectively of the filter media **50B**.

(60) Various configurations for the channels **66, 68** are possible. In the embodiment shown, the

inner channel **66** can be formed by a depending wall **70** of the upper housing **42B** and a first wall **72** of the lower housing **44B** that is spaced radially outwardly from the upper housing wall **70**. The outer channel **68** can be formed by a second wall **74** of the lower housing **44B** and a third wall **76** of the lower housing **44B** that is spaced radially outwardly from the second wall **74**.

(61) The filter media **50B** can be disposed between the first and second walls **72, 74**. The filter media **50** can substantially fill the open space between the first and second walls **72, 74** in the lower housing **44B** and between the upper and lower mesh **46B, 48B**.

(62) FIG. **8A** is a cross-sectional view through the top portion **26** of the air purifier **10** and the air purifier accessory **40B**. A groove **54** in the top portion **26** can function as a seat for the accessory **40B**. The accessory **40B** can have a corresponding extending portion **56B** that is at least partially received in the groove **54**. The groove **54** can be disposed inwardly of the lip or ledge **28**. In other embodiments, the groove **54** can be provided in the accessory **40B** and the extending portion **56B** can be provided on the air purifier **10**. Yet other physical features for aligning the accessory **40B** on the air purifier **10** are possible.

(63) FIGS. **11-13** show an air purifier accessory **40C** according to another aspect of the disclosure. The air purifier accessory **40C** is substantially similar to that of FIGS. **1-7**, and like elements are referred to with the same reference numeral bearing a letter “C.” The accessory **40C** has a dual-layer filter media **50C** between the upper and lower housings **42C, 44C**. The filter media **50C** includes an upper filter layer **78** and a lower filter layer **80** that can be fused together or separate from each other. The layers **78, 80** can contain the same or different filter materials. In one embodiment, the upper layer **78** is a Zeolite filter and the lower layer **80** is a carbon filter.

(64) While not shown, mesh can be disposed on the upper side of the filter media **50C**, on the lower side of the filter media **50C**, in between the layers **78, 80** of the filter media **50C**, or any combination thereof.

(65) The accessory **40C** includes at least one air channel that allows air exiting the air purifier **10** (FIG. **1**) to bypass the dual-layer filter media **50C** so as to not suffocate the air purifier **10**. In the embodiment shown, an inner channel **82** and an outer channel **84** are provided and are disposed radially inwardly and radially outwardly, respectively of the filter media **50C**.

(66) FIGS. **14-16** show an air purifier accessory **40D** according to another aspect of the disclosure. The air purifier accessory **40D** is substantially similar to that of FIGS. **1-7**, and like elements are referred to with the same reference numeral bearing a letter “D.” The accessory **40D** includes a single housing or frame **86** with a press-fit insert **88** comprising the filter media **50D**. The insert **88** can, for example be cardboard or another recyclable, biodegradable and/or compostable material, or combination of materials. Once the filter media **50D** is fully saturated, the insert **88** can be removed from the frame **86** and replaced with a new insert.

(67) The insert **88** can include a filter media matrix **92** defining a plurality of cells **94**. In one embodiment, the matrix **92** can have a honeycomb configuration, with the cells **94** having a hexagonal shape. Other configurations for the matrix **92** and shape for the cells **94** are possible.

(68) The filter media **50D** can be disposed in at least some, alternatively all, of the cells **94**. In one embodiment, at least some of the cells **94** can be empty, e.g. can lack filter media **50D**, in order to allow some air to bypass the accessory **40D** and limit air flow restriction. It is noted that in FIG. **14**, only a portion of filter media **50D** is shown for illustration purposes. The filter media **50D** is not depicted in FIGS. **15-16** for clarity.

(69) The matrix **92** be a plastic component manufactured via injection molding. After injection molding, the filter media **50D** is added to the matrix **92**, for example by filling at least some, alternatively all, cells **94** with a granulate, particulate, or pelletized filter media. Mesh **46D, 48D** can be disposed on the upper and lower sides of the insert **88** to cover the matrix **92** and contain the filter media **50D** within the matrix **92**. The mesh **46D, 48D** is not depicted in FIGS. **15-16** for clarity.

(70) Optionally, the filter media **50D** can include a replacement indicator **90**. The replacement

indicator can be disposed in one cell **94** of the matrix **92** or elsewhere on the insert **88**, and changes color to indicator when to replace the insert **88**. Advantageously, by positioning the replacement indicator **90** on the upward facing surface of the accessory **40D**, a user can readily observe the color when passing by the air purifier, and is notified when replacement of the insert **88** is indicated. It is understood that a similar replacement indicator **90** can be applied to any embodiment of filter media or air purifier accessory disclosed herein.

(71) In one embodiment, the replacement indicator **90** can include a media including a color-shifting chemisorptive coating commercially available as Colorfil™, and changes color as it captures chemicals and odors.

(72) FIGS. **17-18** show an air purifier accessory **40E** according to another aspect of the disclosure. The air purifier accessory **40E** is substantially similar to that of FIGS. **14-16**, and like elements are referred to with the same reference numeral bearing a letter “E.” In this embodiment, the air purifier accessory **40E** includes a one-piece frame **96** having the filter media matrix **92E**. The frame **96** may be a plastic component manufactured via injection molding, e.g. with the matrix **92E** and cells **94E** integrally formed by the molding process. After injection molding, a filter media **50E** is added to the matrix **92E**, for example by filling at least some, alternatively all, cells **94E** with a granulate, particulate, or pelletized filter media. Mesh (not shown) can be disposed on the upper and lower sides of the frame **96** to cover the matrix **92E** and contain the filter media **50E** within the matrix **92E**. It is noted that in FIG. **17**, only a portion of filter media **50E** is shown for illustration purposes. The filter media **50E** is not depicted in FIG. **18** for clarity.

(73) FIGS. **19-22** show the air purifier **10** including an air purifier accessory **40F** according to another aspect of the disclosure. Like elements are referred to with the same reference numeral bearing a letter “F.” Like previous embodiments, the accessory **40F** is disposed on the housing **12** to filter, clean, and/or purify air flowing out of the air **30**. However, rather than flowing axially out of the accessory **40F**, air flows radially out of the accessory **40F**. For example, referring to FIG. **1**, air flowing out of air outlet **30** in a generally axial direction relative to the longitudinal axis X of the housing **12** passes out of the accessory **40** in substantially the same generally axial direction. Referring to FIG. **19**, air passes out of the accessory **40F** in a generally radial direction R relative to the axis X of housing **12** or an axis of the accessory **40F**, which may, in some embodiments, be collinear with axis X. In being “generally axial,” the direction can deviate up to 5 degrees from axis X, alternatively up to 10 degrees, alternatively up to 20 degrees. In being “generally radial,” the direction is about 90 degrees from axis X, alternatively 90±5 degrees, alternatively 90±10 degrees, alternatively 90±20 degrees.

(74) Referring additionally to FIGS. **21-22**, the accessory **40F** includes a frame **98** holding a filter media **50F** and an air deflector **100**. The frame **98** can include an annular body formed by a filter media matrix **104** defining a plurality of cells **106**. The filter media **50F** can be disposed in at least some, alternatively all, of the cells **106**. Mesh **46F**, **48F** can be disposed on the outer and inner sides of the matrix **104** to contain the filter media **50** within the matrix **104**. It is noted that in FIG. **21**, only a portion of filter media **50F** is shown for illustration purposes. The mesh **46F**, **48F** is not depicted in FIGS. **19-20** and **22** for clarity.

(75) In one embodiment, the matrix **104** can have a honeycomb configuration, with the cells **106** having a hexagonal shape. Other configurations for the matrix **104** and shape for the cells **106** are possible. The cells **106** can be oriented on an angle as shown, e.g. which can simplify manufacturing in the case of an injection-molded frame **98**. In other embodiments, the cells **106** can be orientated radially, upwardly, downwardly, or a combination thereof.

(76) It is noted that the configuration of the flow diverters of the outlet of the air purifier **10** can, in one embodiment, create a vortex-like flow of air, and the shape of the matrix **104** can be configured to follow the vortex-like flow of air, for example with the cells **106** arranged in a spiral pattern and individual cells **106** pointing upward. The plurality of cells **106** can be generally uniform in size, shape, and/or orientation, or may have a variety of sizes, shapes, and/or orientation.

(77) Optionally, the accessory **40F** can include a base ring **108** on a bottom side of the frame **98**. The base ring **108** is used to install the accessory **40F** on the air purifier **10**, and can include or more physical features for aligning the accessory **40F** on the air purifier **10**. In one embodiment, the base ring **108** includes a groove **110** (FIG. 22) that fits over the lip or ledge **28** in the top portion **26** to nest the accessory **40F** on top of the air purifier **10**. Other physical features for aligning the accessory **40F** on the air purifier **10** are possible.

(78) The air deflector **100** is primarily disposed within the interior of the frame **98** and defines a closed area **112** in which air does not flow. The air deflector **100** is spaced from an inner side **114** of the matrix **104** to define an open area **116**, which can extend circumferentially around the air deflector **100**. The air exiting the air purifier **10** flows into the open area **116** between the air deflector **100** and the inner side **114** of the matrix **104**, passes through the cells **106**, and then out of an outer side **118** of the matrix **104**.

(79) The accessory **40F** includes at least one air inlet **150** on an inflow side **152** of the frame **98** and at least one air outlet **154** on an outflow side **156** of the frame **98**. In the embodiment of FIGS. 19-22, the accessory **40F** has an air inlet **150** formed on a lower side of the frame **98**, the lower side defining the inflow side **152**. The accessory **40D** includes a plurality of air outlets **154** on the outflow side **156** of the frame **98**, and the outer side **118** of the matrix **104** can form the outflow side **156**.

(80) The air deflector **100** is configured to deflect the air to be filtered as it flows through the frame **98**, with air flowing through the open area **116**. Air exiting the air purifier **10** flows axially into the open area **116** through the inflow side **152** of the frame **98** and flows radially out of the open area **116** through the outflow side **156** of the frame **98**.

(81) In one embodiment, the air deflector **100** is defined by a trumpeted center section **120**. The trumpeted center section **120** has a lower end **122** and an upper end **124**, where the upper end **124** has a larger diameter than the lower end **122**. This provides the open area **116** with a larger cross-sectional area at the lower end of the air deflector **100** and a smaller cross-sectional area at the upper end of the air deflector **100**. The upper end **124** can include a flange **126** disposed on a top side of the frame **98**, with the trumpeted center section **122** extending downward from the top side into the interior of the annular body **102**. The trumpeted center section **120** is closed to air flow, such that air that enters the open area **116** flows out through the frame **98**.

(82) The lower end **122** of the air deflector **100** can define the air inlet **150** on an outer side thereof. The air inlet **150** can, for example, be formed by the open space between the lower end **122** and the base ring **108**, as best seen in FIG. 22. The air inlet **150** can extend circumferentially around the lower end **122** of the air deflector **100**. In other embodiments, the air inlet **150** can extend less than the entire circumference of the air deflector **100** and/or multiple air inlets can be provided.

(83) The presence of the air deflector **100** can affect the speed of air as it passes through the accessory **40F**. In particular, air moves faster near the surface of the air deflector **100** than it does farther away from the air deflector **100**. With a trumpeted shape, air speed is higher at the upper end **124** than at the lower end **122**.

(84) The interior of the air deflector **100** can be hollow to provide access to a user interface UI **36** of the air purifier **10**. The lower end **122** can have an opening sized to fit around the user interface, such that a center area of the accessory **40F** is clear and the user interface is accessible when the accessory **40F** is present on the air purifier **10**.

(85) Optionally, the accessory **40F** can include at least one bypass outlet that allows air exiting the air purifier **10** (FIG. 1) to bypass the filter media **50F** so as to not suffocate the air purifier **10**. In the embodiment shown, multiple bypass outlets **128** are provided in the frame **98** and are disposed about the circumference of the frame **98**.

(86) The cells **106** and bypass outlets **128** can collectively define the outlet area through which air can leave the accessory **40**. The bypass outlets **128** define a small percentage of the outlet area, with the cells **106** defining a much larger percentage of the outlet area. Accordingly, the majority of

air that enters the open area **116** flows out of the air purifier **10** through filtration media **50F** via the cells **106**, while a minority bypasses the filtration media **50F** via the bypass outlets **128**.

(87) Various configurations for the bypass outlets **128** are possible. In one embodiment, the bypass outlets **128** can comprise narrow slits in the frame **98**. The slits can be elongated horizontally, and are relatively short vertically.

(88) The bypass outlets **128** may preferably be disposed at the upper end of the frame **98**, where airflow velocity is highest due to the trumpeted shape of the air deflector **100**. For example, the bypass outlets **128** can be disposed close to the surface of the deflector **100**, at the wider upper end **124**. The bypass outlets **128** can be above the matrix **104** and below the top side of the frame **98**. Other locations for the bypass outlets **128** are possible.

(89) The frame **98** may be a plastic component manufactured via injection molding, e.g. with the matrix **104** and cells **106**, and optionally the bypass channels **128** integrally formed by the molding process. After injection molding, a filter media **50F** is added to the matrix **104**, for example by filling at least some, alternatively all, cells **106** with a granulate, particulate, or pelletized filter media. Mesh **46F**, **48F** (FIG. **21**) can contain the granulate, particulate, or pelletized filter media within the matrix **104**. Before or after adding the filter media **50F**, the air deflector **100** and optional base ring **108** are assembled on the frame **98**.

(90) The frame **98** may define the longitudinal axis X of the accessory **40F**, with the axis X extending through the open center of the frame **98**. The axis X may also extend through the open center of the air deflector **100**.

(91) In some embodiments, the frame **98** can be formed in multiple pieces and assembled together in an annular ring shape as shown. To assemble the pieces of the frame **98**, the air deflector **100**, and the base ring **108**, a suitable joining method such as adhesion, gluing, bonding, welding, and the like, is used. In one example, as shown in FIG. **21**, the frame **98** is formed in three sections **98A**, **98B**, **98C**.

(92) With a multi-piece frame, each section **98A**, **98B**, **98C** can be filled, or partially filled, with filter media **50F** before assembling the sections together, which can result in a greater volume of filter media **50F** in the frame **98**. Each section **98A**, **98B**, **98C**, which a section of outer mesh **46F** applied, can be laid on its outer side, filled with filter media **50F**, and then closed by a section of inner mesh **48F**. The filled sections **98A**, **98B**, **98C** are then assembled together to complete the annular frame **98**. The air deflector **100** can be installed through the top side of the frame **98** and attached to the upper edge thereof.

(93) FIG. **23** show the air purifier **10** including an air purifier accessory **40G** according to another aspect of the disclosure. The accessory **40G** is substantially similar to the accessory **40F** of FIGS. **19-22**, and like elements are referred to with the same reference numeral bearing a letter "G." The accessory **40G** differs in that, rather than being closed to air flow, the air deflector **100G** has at least one open area **130** through which ambient air A, which has not passed through the air purifier **10**, can enter the accessory **40G** and be filtered, providing additional odor control.

(94) Various configurations for the open area **130** are possible. In one embodiment, the air deflector includes an upper section **132** and a lower section **134**, and the open area **130** is defined between the upper and lower sections **132**, **134**. The lower section **134** can have a top edge **136** that flares outwardly beyond a lower edge **138** of the upper section **132** to create a gap **140** through which air can pass. The gap **140** can extend substantially around a circumference of the air deflector **100G**. Ribs **142** can connect the upper and lower sections **132**, **134** together and can cross the gap **140**.

(95) The air deflector **100G** may be a plastic component manufactured via injection molding, e.g. with the upper and lower sections **132**, **134** integrally formed by the molding process. In another embodiment, the air deflector **100G** be formed in multiple pieces and assembled together.

(96) During operation of the air purifier **10**, high velocity air moving on the outside of the air deflector **100G** creates a negative pressure zone, which sucks ambient air A down through the open area **130** in the deflector **100G**. The ambient air A flows through the open area **130** between the

upper and lower sections **132**, **134** of the air deflector **100G** and is filtered by passing through filtration material **50G** via the cells **106**.

(97) In other embodiments, the open area **130** can be provided in various other locations and configurations on the air deflector **100G**. For example, the open area **130** can be formed as a plurality of perforations, slots, or other apertures in the deflector **100**.

(98) The terms “comprising” or “comprise” are used herein in their broadest sense to mean and encompass the notions of “including,” “include,” “consist(ing) essentially of,” and “consist(ing) of.” The use of “for example,” “e.g.,” “such as,” and “including” to list illustrative examples does not limit to only the listed examples. Thus, “for example” or “such as” means “for example, but not limited to” or “such as, but not limited to” and encompasses other similar or equivalent examples.

(99) The above description relates to general and specific embodiments of the disclosure. However, various alterations and changes can be made without departing from the spirit and broader aspects of the disclosure as defined in the appended claims, which are to be interpreted in accordance with the principles of patent law including the doctrine of equivalents. As such, this disclosure is presented for illustrative purposes and should not be interpreted as an exhaustive description of all embodiments of the disclosure or to limit the scope of the claims to the specific elements illustrated or described in connection with these embodiments. Any reference to elements in the singular, for example, using the articles “a,” “an,” “the,” or “said,” is not to be construed as limiting the element to the singular.

Claims

1. An air purifier assembly, comprising: an air purifier comprising: a housing having a longitudinal axis; an air flow path through the housing including an air inlet and an air outlet; a blower disposed within the housing, the blower defining a portion of the air flow path and adapted to pull air in through the air inlet and to push air out of the air outlet; and an air treatment device disposed within the housing to treat air flowing through the air flow path; an air purifier accessory disposed on the air outlet, outside of the housing, to treat air flowing out of the air outlet, the air purifier accessory comprising: a frame; a filter media within the frame for filtering air discharged from the air purifier; at least one air inlet on an inflow side of the frame; a plurality of air outlets on an outflow side of the frame, wherein the outflow side of the frame is perpendicular to the inflow side of the frame; and an air deflector configured to deflect the air to be filtered as it flows through the frame, the air deflector defining a closed area within the frame in which air does not flow and an open area within the frame in which air flows; wherein air exiting the air purifier flows axially into the open area through the inflow side of the frame and flows radially out of the open area through the outflow side of the frame.
2. The air purifier assembly of claim 1, wherein the housing defines a lower portion and an upper portion, the lower portion adapted to rest on a surface, wherein the air outlet is formed in the upper portion of the housing and the air purifier accessory is adapted to rest on the upper portion of the housing to cover the air outlet.
3. The air purifier assembly of claim 1, wherein the housing comprises a cylindrical body and the air purifier accessory comprises an annular body formed by a filter media matrix defining a plurality of cells, wherein the filter media is disposed in at least some of the plurality of cells.
4. The air purifier assembly of claim 3, wherein the air deflector is spaced from an inner side of the filter media matrix to define the open area, wherein the air exiting the air purifier flows into the open area between the air deflector and the inner side of the filter media matrix.
5. The air purifier assembly of claim 3, wherein the filter media matrix comprises a honeycomb configuration and the plurality of cells comprise a hexagonal shape.
6. The air purifier assembly of claim 1, wherein the air purifier comprises a user interface, and the air purifier accessory comprises an access opening to allow access to the user interface, wherein the

air deflector defines the access opening.

7. The air purifier assembly of claim 1, wherein the air purifier accessory comprises at least one bypass outlet provided in the frame, wherein the bypass outlet allows air discharged from the air purifier to bypass the filter media.

8. The air purifier assembly of claim 7, wherein the air deflector comprises a trumpeted center section having a lower end and an upper end, where the upper end has a larger diameter than the lower end, and wherein the at least one bypass outlet is disposed at an upper end of the frame.

9. The air purifier assembly of claim 7, wherein the air deflector includes an upper end and a lower end, and wherein the open area has a larger cross-sectional area at the lower end of the air deflector and a smaller cross-sectional area at the upper end of the air deflector, wherein the at least one bypass outlet is disposed at the upper end of the air deflector.

10. An air purifier accessory, comprising: a frame adapted to be coupled to an exterior of an air purifier and defining an longitudinal axis; a filter media within the frame for filtering air discharged from the air purifier; at least one air inlet on an inflow side of the frame; a plurality of air outlets on an outflow side of the frame, wherein the outflow side of the frame is perpendicular to the inflow side of the frame; and an air deflector configured to deflect the air to be filtered as it flows through the frame, the air deflector defining an open area within the frame in which air flows; wherein air exiting the air purifier flows axially into the open area through the inflow side of the frame and flows radially out of the open area through the outflow side of the frame.

11. The air purifier accessory of claim 10, wherein the frame comprises an annular body formed by a filter media matrix defining a plurality of cells, wherein the filter media is disposed in at least some of the plurality of cells.

12. The air purifier accessory of claim 11, comprising a mesh disposed on at least a portion of the filter media matrix to contain the filter media in the filter media matrix.

13. The air purifier accessory of claim 11, wherein the filter media matrix comprises a honeycomb configuration and the plurality of cells comprise a hexagonal shape.

14. The air purifier accessory of claim 11, wherein the air deflector is spaced from an inner side of the filter media matrix to define the open area, wherein the air exiting the air purifier flows into the open area between the air deflector and the inner side of the filter media matrix.

15. The air purifier accessory of claim 10, wherein the air deflector comprises a trumpeted center section having a lower end and an upper end, where the upper end has a larger diameter than the lower end.

16. The air purifier accessory of claim 15, wherein the upper end comprises a flange disposed on a top side of the frame, with the trumpeted center section extending downward from the top side into an interior of the frame.

17. The air purifier accessory of claim 10, wherein the air deflector comprises a lower end and an upper end, and wherein the open area comprises a first cross-sectional area at the lower end of the air deflector and a second, smaller cross-sectional area at the upper end of the air deflector.

18. The air purifier accessory of claim 10, comprising at least one bypass outlet provided in the frame, wherein the bypass outlet allows air discharged from the air purifier to bypass the filter media.

19. The air purifier accessory of claim 18, wherein the air deflector comprises a trumpeted center section having a lower end and an upper end, where the upper end has a larger diameter than the lower end, and wherein the at least one bypass outlet is disposed at an upper end of the frame.

20. The air purifier accessory of claim 10, wherein the air deflector includes an upper section and a lower section, and the open area is defined between the upper and lower sections.

21. The air purifier accessory of claim 20, wherein the lower section comprises a top edge that flares outwardly beyond a lower edge of the upper section to create a gap through which air can pass.
